

L4. Multiple Qubits and Quantum Entanglement

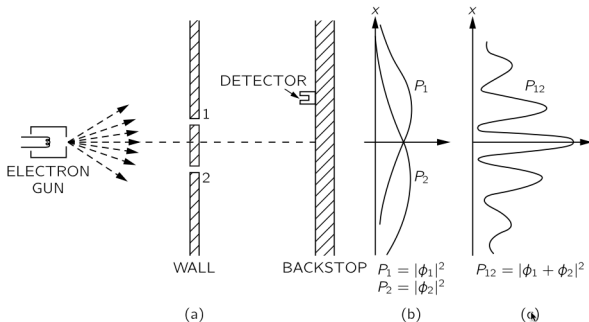
Xin Wan

Zhejiang University

Q Weirdness I: Superposition and Interference

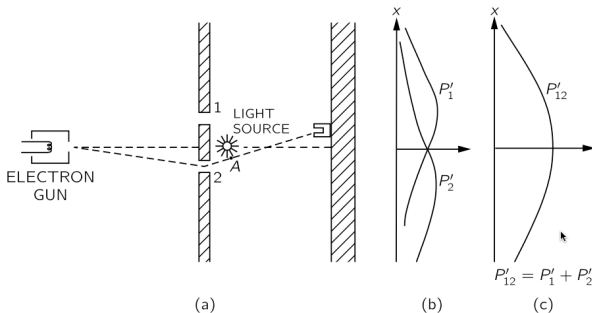
- For a photon polarized along $|p\rangle$ passing through two indistinguishable possible paths ($|h\rangle, |v\rangle$), the probability of measuring polarized photon along $|n\rangle$ is

$$P_{\text{tot}}^{\text{quantum}} = |\langle n|h\rangle\langle h|p\rangle + \langle n|v\rangle\langle v|p\rangle|^2.$$

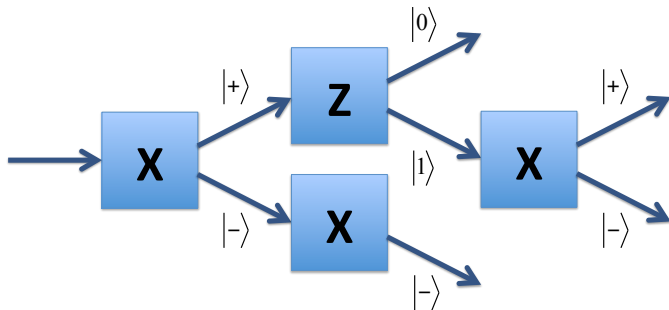


- In the which-way setup, we further determine which of the two paths ($|h\rangle$ or $|v\rangle$) the photon passes. The final total probability of measurement becomes, then,

$$P_{\text{tot}}^{\text{classical}} = |\langle p|h\rangle\langle h|n\rangle|^2 + |\langle p|v\rangle\langle v|n\rangle|^2.$$



Q Weirdness II: Measurement and Uncertainty¹



- Which measurement results in the sequential Stern-Gerlach experiment can be explained by a classical probability theory? Which cannot? **Note $[X, Z] \equiv XZ - ZX \neq 0!$**

¹We are going to discuss entanglement, another Q weirdness in this lecture.

- The uncertainties associated with the observables A and B for any state ψ are bounded by the **uncertainty principle**:

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|.$$

- The statistical spread of the measurement of A around its mean value $\langle A \rangle = \langle \psi | A | \psi \rangle$ is defined as

$$(\Delta A)^2 \equiv \langle A^2 \rangle - \langle A \rangle^2 = \langle \psi | A^2 | \psi \rangle - \langle \psi | A | \psi \rangle^2.$$

- The familiar example for $A \equiv x$ and $B \equiv p_x = -i\hbar \partial / \partial x$ is

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}, \quad \text{as } [x, p_x] = i\hbar.$$

Outline

- Tensor Products
- Entangled States and EPR Paradox
- Measuring Information in a Bell State
- Concept of Density Matrices

4.1 Tensor Products

- A pair of classical bits can have (or be in) any of the four possible states 00, 01, 10, 11.
- If a bit is represented by $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the two-bit states can be represented by four column vectors

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

- They are examples of tensor products.

- In general, the **tensor product** of an $m \times n$ matrix (vector) A and a $p \times q$ matrix (vector) B is an $mp \times nq$ matrix in the form

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

- The tensor product of two vectors $|u\rangle$ and $|v\rangle$ is often abbreviated as $|u\rangle|v\rangle$ or $|uv\rangle$, as was done above.

4.2 Entangled States and EPR Paradox

Take-Home Messages

- In general, multi-qubit states can be divided into separable states and entangled states. The former can be written as a tensor product, while the latter cannot. The Bell states are examples of entangled states.
- In the EPR paradox, no local information is encoded in the entangled pairs, such that the information can travel faster than light upon local measurement.

Combining Two Qubits

- In the case of two qubits the vector space is, therefore, four-dimensional, with an orthonormal basis

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

- The general state $|\Psi\rangle$ that nature allows us to associate with two qubits is any normalized superposition of the four orthogonal classical states,

$$|\Psi\rangle = \alpha_1 |00\rangle + \alpha_2 |01\rangle + \alpha_3 |10\rangle + \alpha_4 |11\rangle$$

with the complex amplitudes being constrained only by the normalization condition $\sum_j |\alpha_j|^2 = 1$.

- In other words, two-qubit states are unit vectors in $(\mathbb{C}^2)^{\otimes 2} = \mathbb{C}^2 \otimes \mathbb{C}^2$.
 - If the first qubit is in $|0\rangle$, and the second qubit is in $|1\rangle$, then the two-qubit state is $|0\rangle \otimes |1\rangle$, $|0\rangle|1\rangle$, or $|01\rangle$.
 - If the first qubit is in $\alpha_1 |0\rangle + \beta_1 |1\rangle$, and the second qubit is in $\alpha_2 |0\rangle + \beta_2 |1\rangle$, then the **product state** is

$$\begin{aligned}
 & (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) \\
 &= \alpha_1 \alpha_2 |00\rangle + \alpha_1 \beta_2 |01\rangle + \beta_1 \alpha_2 |10\rangle + \beta_1 \beta_2 |11\rangle
 \end{aligned}$$

- In the two case, even though we put the two qubits together, the information they encode is still *separable*. If we operate on the first qubit only, the second remains unchanged:

$$(L \otimes I)(|u\rangle \otimes |v\rangle) = (L|u\rangle) \otimes |v\rangle.$$

- It takes 4 real parameters to specify a two-qubit product state (or separable state).
 - 4 real (i.e., 2 complex) parameters for each qubit, minus 2 normalization conditions and 2 overall phase factor.
- On the other hand, it takes 6 real parameters to specify the most general two-qubit state.
 - 8 real (i.e., 4 complex) parameters, minus 1 normalization condition and 1 overall phase factor.
- Obviously, the space of all two-qubit states is much richer than that of the product states, which can be prepared separately for each qubit.

- The other class of states are known as the **entangled states**, exemplified by the famous Bell states. One of the four Bell states is

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle).$$

- In the state particle 1 can be in state $|0\rangle$ and particle 2 in state $|1\rangle$, or vice versa, but there is no way of knowing which particle is in which state. All that is defined is the fact that the two qubits are different.

EPR Paradox

- Entanglement is closely linked to the issue of locality (or non-locality) in quantum theory.
- In particular, if the two particles in an entangled state are widely separated, then a measurement on one will immediately influence the quantum state of the other one. It seems as if the particles are communicating faster than the speed of light.
- Because of this, Einstein and coworkers challenged the idea that quantum mechanics could provide a complete description of reality [Einstein, Podolsky, and Rosen, *Phys. Rev.* **47**, 777 (1935)].

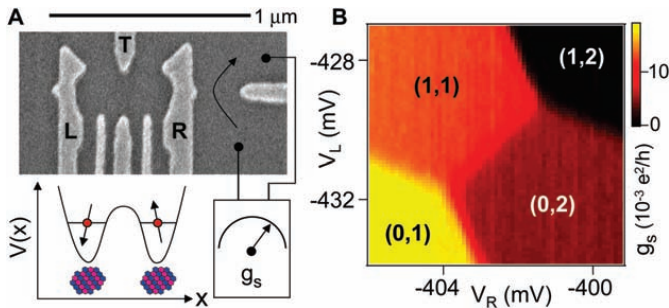
- In Bohm's version of the EPR paradox, the two particles are in the Bell state

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle).$$

- One can measure the first qubit according to the following measurement rule

$$|\Psi\rangle = \left. \begin{array}{l} \alpha_0|0\rangle|\Phi_0\rangle \\ + \alpha_1|1\rangle|\Phi_1\rangle \end{array} \right\} \begin{array}{c} \text{---} \boxed{\overset{x}{M}} \text{---} \\ \text{---} \end{array} \begin{array}{l} |x\rangle \\ |\Phi_x\rangle \end{array} \quad p = |\alpha_x|^2$$

Preparing a Bell State in Quantum Dots

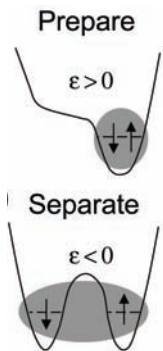


- Double quantum dots can be prepared on the surface of a 2D electron gas with electrostatic gates. Charge states (m,n) can be tuned by tuning gate voltages V_L and V_R .

- Tight confinement in $(0, 2)$ favors a Bell-state configuration (due to Pauli exclusion principle)

$$|\psi_{-}\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle).$$

- Controlled voltage pulses can switch the ground state to $(1, 1)$, separating the two electron spins in the two neighboring quantum dots.
- After they are separated, the two spins remain entangled.



- Upon the measurement of particle 1, one knows the state of the faraway particle 2 instantaneously.
- However, the special theory of relativity is not violated because no information travels faster than light (local in the quantum field theoretical sense).
 - Before the measurement, particle 2 has equal probabilities in states $|0\rangle$ and $|1\rangle$. This remains to be true (in terms of an external observer's statistical model) after the measurement.
 - The measurement of particle 1 does not *a priori* fix the state of particle 2; there is no information transfer.

4.3 Measuring Information in a Bell State

Take-Home Messages

- The expectation value of a measurement L of a system in state $|\psi\rangle$ is $\langle L \rangle \equiv \langle \psi | L | \psi \rangle$. It is the statistical average of many measurements on an ensemble of identical states.
- One cannot identify any spatial direction $\hat{n}$, along which a single spin in a Bell state has nonzero expectation value, i.e., $\langle \vec{\sigma} \cdot \hat{n} \rangle \neq 0$.

Expectation Values

- The two-particle Bell state cannot be written as the tensor product of single-particle states. The single-particle states are *entangled*, not separable, in such superpositions.
- This means that all of the information is distributed among two qubits (i.e., not local), and that none of the individual systems carries any information. This is the essence of *entanglement* and is one of the novel and counterintuitive features of quantum mechanics.
- For further illustration, we introduce the concept of expectation values, or average values.

- Suppose we consider an experiment that measures an observable L , whose eigenvalues are λ with corresponding eigenvectors $|\lambda\rangle$, i.e., $L|\lambda\rangle = \lambda|\lambda\rangle$.
- Suppose that the normalized state of a quantum system is ψ , which can be expanded in the orthonormal basis $\{|\lambda\rangle\}$:

$$|\psi\rangle = \sum_{\lambda} |\lambda\rangle \langle \lambda | \psi \rangle.$$

- Let L operate on ψ :

$$L|\psi\rangle = \sum_{\lambda} L|\lambda\rangle \langle \lambda | \psi \rangle = \sum_{\lambda} \lambda |\lambda\rangle \langle \lambda | \psi \rangle.$$

- The expectation value of L of the system is defined as

$$\langle L \rangle \equiv \langle \psi | L | \psi \rangle = \sum_{\lambda} \lambda \langle \psi | \lambda \rangle \langle \lambda | \psi \rangle = \sum_{\lambda} \lambda P(\lambda).$$

- This is the weighted sum of L 's eigenvalues λ with probability $P(\lambda) = \langle \psi | \lambda \rangle \langle \lambda | \psi \rangle$.
- Suppose we perform a large number of identical experiments. We can identify $P(\lambda)$ as the percentage of observations whose result is λ .
- In the statistical sense, the experimental average is identical to the mathematical definition of $\langle L \rangle$.

Information in a Bell State

- For a one-qubit state $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$, one can always find a particular direction $\hat{n}$ for which $\langle \vec{\sigma} \cdot \hat{n} \rangle = 1$.
 - One can check, $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ is the direction.
 - This still holds when we measure one of the qubit in a two-qubit product state, such as $|\psi\rangle \otimes |\psi'\rangle$.
- However, if one measures the first qubit in the Bell state $|\Psi_+\rangle$, the result is $\langle \vec{\sigma}^{(1)} \cdot \hat{n} \rangle = 0$ for any $\hat{n}$.
 - Let us operate $\sigma_n^{(1)} = \vec{\sigma}^{(1)} \cdot \hat{n}$, or more precisely, $(\vec{\sigma}^{(1)} \cdot \hat{n}) \otimes I$:
$$\sigma_n^{(1)} |\Psi_+\rangle \propto (\cos \theta |0\rangle + e^{i\phi} \sin \theta |1\rangle) |1\rangle + (e^{-i\phi} \sin \theta |0\rangle - \cos \theta |1\rangle) |0\rangle.$$
 - A quick comparison with $|\Psi_+\rangle \propto |01\rangle + |10\rangle$ shows that $\langle \Psi_+ | \vec{\sigma}^{(1)} \cdot \hat{n} | \Psi_+ \rangle = 0$, regardless of the direction of $\hat{n}$.

- If the expectation value of a component of $\vec{\sigma}^{(1)}$ vanishes, it means that the experimental outcome is equally likely to be ± 1 . In other words, the outcome is completely uncertain.
- Assume Charlie prepare two qubits in the entangled state and send one qubit to Alice and one to Bob without looking at (i.e., measuring) the qubits. Although Alice and Bob know exactly what the combined system is in, they can predict nothing about the outcome of their individual measurements. This is the true weirdness of entanglement, which so disturbed Einstein.
- What operator, if any, measures the information of the entangle state?

- In Lab 3 Exercise E2 we have constructed the dot product of the vector-operator $\vec{\sigma}^{(1)}$ and $\vec{\sigma}^{(2)}$, which is manifestly rotational invariant,

$$\vec{\sigma}^{(1)} \cdot \vec{\sigma}^{(2)} = \sigma_x^{(1)} \sigma_x^{(2)} + \sigma_y^{(1)} \sigma_y^{(2)} + \sigma_z^{(1)} \sigma_z^{(2)}.$$

This operator always yields a certain information:

$$\langle \Psi_+ | \vec{\sigma}^{(1)} \cdot \vec{\sigma}^{(2)} | \Psi_+ \rangle = \text{const.}$$

- What is the constant? Is it possible that Alice and Bob can measure with Stern-Gerlach apparatuses to evaluate $\vec{\sigma}^{(1)} \cdot \vec{\sigma}^{(2)}$?

4.4 Concept of Density Matrices

Take-Home Messages

- Density matrices can be regarded as the quantum generalization of classical probabilities.
 - The density matrix is $\rho \equiv |\psi\rangle\langle\psi|$ for a pure state $|\psi\rangle$, and $\rho = \sum_i P_i |\psi_i\rangle\langle\psi_i|$ for a mixed state.
 - The expectation value, or averaged value, of measurement L on a state with density matrix ρ is $\langle L \rangle = \text{Tr} \rho L$. This applies to both pure states and mixed states.

- The reduced density matrix for a subsystem A , defined as $\rho_A = \text{Tr}_B \rho$, allows us to calculate with variables only in A . For example, the expectation value of operator L_A defined in A is

$$\langle L_A \rangle = \text{Tr}_A(\rho_A L_A).$$

- The entanglement between two subsystems can be detected by the bipartite entanglement entropy, which is defined as

$$S_A \equiv -\text{Tr}_A(\rho_A \log_2 \rho_A).$$

Density Matrices

- A complete description of the Bell state involves the variables both at Alice and Bob's ends. To evaluate the result of a measurement to be carried out only at, say, Alice's end, we still operate on the whole state, which involves both ends.
- This is counterintuitive. Do we have to know the whole state? Is there a way to get rid of Bob's variables?
- The answers are yes. But, to capture the complete information Alice (not Bob) has, we need to introduce new tools: density matrix and reduced density matrix.

- We begin our introduction to density matrix with an alternative expression for evaluating the expectation value.
- First, let us define the **trace** of an operator L to be

$$\text{Tr}L = \sum_i \langle i|L|i\rangle,$$

where $\{|i\rangle\}$ is a complete set of basis vectors.

- The trace of a projection operator is 1.
- The trace of a Hermitian operator is the sum of its eigenvalues.
- Recall the completeness condition $\sum_i |i\rangle\langle i| = I$,

$$L = LI = \sum_i L|i\rangle\langle i|.$$

- The expectation value of L in state $|\psi\rangle$ can be, alternatively, written as

$$\langle\psi|L|\psi\rangle = \sum_i \langle\psi|L|i\rangle \langle i|\psi\rangle = \sum_i \langle i|\psi\rangle \langle\psi|L|i\rangle,$$

where we simply reverse the order of the two numbers in the second equality.

- According to the definition $\text{Tr}L = \sum_i \langle i|L|i\rangle$,

$$\langle\psi|L|\psi\rangle = \text{Tr}|\psi\rangle\langle\psi|L \equiv \text{Tr}\rho L.$$

$\rho \equiv |\psi\rangle\langle\psi|$ is the **density matrix** of the pure state $|\psi\rangle$.

- Density matrix can also be used to describe a state of incomplete information.
- For example, Charlie has prepared a qubit and sends it to Alice. He does not tell her along which axis the qubit was prepared. Instead, he tells her that the state is equally likely to be $|0\rangle$ along the z axis or $|r\rangle$ along the x axis. Remember the protocol of quantum key distribution?
- Alice can reason that the expectation value of L is

$$\langle L \rangle = \frac{1}{2} \langle 0 | L | 0 \rangle + \frac{1}{2} \langle r | L | r \rangle = \frac{1}{2} \text{Tr} |0\rangle \langle 0| L + \frac{1}{2} \text{Tr} |r\rangle \langle r| L.$$

- Now we can define a density matrix ρ that encodes Alice's knowledge such that $\langle L \rangle = \text{Tr} \rho L$.
- One can read that the density matrix ρ is half the projection operator onto $|0\rangle$ plus half the projection operator onto $|r\rangle$

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|r\rangle\langle r|,$$

which represents a **mixed state** (with $1/2$ being classical probability).

- One can generalize $\rho = \sum_i P_i |\psi_i\rangle\langle\psi_i|$ to be a mixture of projection operators with probabilities P_i .

Reduced Density Matrix

- Assume the third party Charlie prepares two qubits (!) in a quantum state $|\psi\rangle$ and, then, sends Alice and Bob one qubit each.
- Without knowing the state $|\psi\rangle$ a priori, Alice attempts to measure a local operator L_A to gain information.
- We assume that the process repeats, such that Alice can obtain the expectation value of L_A by calculating the averaged measurement result. Can we define a density matrix for Alice's measurement such that $\langle L_A \rangle = \text{Tr} \rho_A L_A$? How is it related to $\rho = |\psi\rangle\langle\psi|$?

- Let Charlie prepare a separable state $|\psi\rangle = |\lambda\rangle_A |\phi\rangle_B$, whose density matrix is

$$\rho = |\psi\rangle\langle\psi| = |\lambda\rangle\langle\lambda| \otimes |\phi\rangle\langle\phi|.$$

- The averaged measurement result of L_A (while Bob does nothing) is

$$\begin{aligned} \langle L_A \rangle &\equiv \langle L_A \otimes I_B \rangle = \text{Tr} \rho (L_A \otimes I_B) \\ &= \sum_{i_A, j_B} \langle i_A | \langle j_B | \rho (L_A \otimes I_B) | i_A \rangle | j_B \rangle \\ &= \left[\sum_{i_A} \langle i_A | \lambda \rangle \langle \lambda | L_A | i_A \rangle \right] \left[\sum_{j_B} \langle j_B | \phi \rangle \langle \phi | I_B | j_B \rangle \right] \end{aligned}$$

- The second bracket is identity, as

$$\sum_{j_B} \langle j_B | \phi \rangle \langle \phi | I_B | j_B \rangle = \sum_{j_B} \langle \phi | j_B \rangle \langle j_B | \phi \rangle = \langle \phi | \phi \rangle = 1.$$

- We arrive at

$$\langle L_A \rangle = \sum_{i_A} \langle i_A | (|\lambda\rangle \langle \lambda| L_A) | i_A \rangle \equiv \text{Tr}_A(\rho_A L_A),$$

where we define the **reduced density matrix** as

$$\rho_A = \text{Tr}_B \rho = |\lambda\rangle \langle \lambda|.$$

- Therefore, knowing ρ_A is sufficient to evaluate $\langle L_A \rangle$, i.e., to describe Alice's measurement results. The subsystem is also in a pure state, such that $\text{Tr} \rho_A^2 = \text{Tr} \rho_A = 1$.

- Next, consider the case when Alice and Bob share a pair of entangled qubits, e.g., in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle).$$

- We can generalize the concept of reduced density matrix for the entangled state as

$$\rho_A = \text{Tr}_B |\psi\rangle\langle\psi| = \sum_j \langle j_B|\psi\rangle \langle\psi|j_B\rangle,$$

where $\{|j_B\rangle\}$ is an orthonormal basis set for Bob's qubit.

- The reduced density matrix satisfies the following properties:
 - $\text{Tr}_A \rho_A = 1$, as $\text{Tr} \rho = \text{Tr}_A \text{Tr}_B \rho = 1$.
 - As in the separable case, knowing ρ_A is sufficient to evaluate $\langle L_A \rangle = \text{Tr}_A(\rho_A L_A)$.
 - One can obtain an orthonormal set $\{|\lambda_i\rangle_A\}$ and the corresponding eigenvalues $\{\lambda_i\}$ by diagonalizing ρ_A . They are uniquely associated with the eigenstates $\{|\phi_i\rangle_B\}$ with the identical eigenvalues $\{\lambda_i\}$ of $\rho_B = \text{Tr}_A |\psi\rangle\langle\psi|$.
 - Indeed, one can decompose (i.e., the Schmidt decomposition)

$$|\psi\rangle = \sum_i \sqrt{\lambda_i} |\lambda_i\rangle_A |\phi_i\rangle_B.$$

- However, in the entangled case $\text{Tr}\rho_A^2 = \text{Tr}\rho_B^2 \neq 1$. In other words, the subsystem is described by a mixed-state reduced density matrix.
- As we discussed in Lecture 1, the von Neumann (Shannon) entropy measures the amount of information. Likewise, we can define **entanglement entropy** to detect the bipartite entanglement between A and B subsystems as

$$S_A \equiv -\text{Tr}_A(\rho_A \log_2 \rho_A) = -\sum_i \lambda_i \log_2 \lambda_i$$

to measure the entanglement of the two subsystems quantitatively. Notice that $S_A = S_B$.

Summary

- Einstein and his collaborators did point out, in the EPR paradox, one of the strangest things in quantum mechanics. But unlike what they challenged, the current understanding of quantum mechanics is complete. There are no hidden variables.
- For a multi-qubit system, information is encoded in the tensor space in richer and nonlocal ways. The entanglement among the qubits is a resource that is crucially useful in quantum information and computation.

- Density matrices are more general than wave functions, as they can describe both pure and mixed states. The key properties of density matrices are:
 - Density matrices are Hermitian $\rho_{ij} = \rho_{ji}^*$.
 - The trace of a density matrix is 1, $\text{Tr}(\rho) = 1$.
 - The eigenvalues of the density matrix are all positive and lie between 0 and 1.
 - For a pure state,

$$\rho^2 = \rho, \quad \text{Tr}(\rho^2) = 1.$$

- For a mixed state (or a subsystem of an entangled state)

$$\rho^2 \neq \rho, \quad \text{Tr}(\rho^2) < 1.$$

References

- L. Susskind and A. Friedman, Quantum Mechanics (Basic Books, 2014), Chaps. 6-7.
- M. Nakahara and T. Ohmi, Quantum Computing: From Linear Algebra to Physical Realizations (CRC Press, 2008), Chaps. 1-3.

Homework

P4-1. Suppose

$$\rho = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{3}{4} \end{pmatrix}.$$

Calculate ρ^2 , $\text{Tr}\rho$, and $\text{Tr}\rho^2$. Can ρ be a density matrix? If yes, does it represent a pure state or a mixed state?

P4-2. Consider two qubits in an entangled state

$$|\Psi\rangle = \sqrt{3}|10\rangle - \sqrt{2}|01\rangle.$$

Normalize the state. Calculate the density matrix ρ . Calculate the following expectation values: $\langle\sigma_x^{(1)}\rangle$, $\langle\sigma_y^{(1)}\rangle$, $\langle\sigma_z^{(1)}\rangle$, $\langle\sigma_x^{(2)}\rangle$, $\langle\sigma_y^{(2)}\rangle$, $\langle\sigma_z^{(2)}\rangle$, $\langle\sigma_z^{(1)}\sigma_z^{(2)}\rangle$, and $\langle\sigma_x^{(1)}\sigma_x^{(2)}\rangle$. Calculate the correlation

$$\langle\sigma_z^{(1)}\sigma_z^{(2)}\rangle - \langle\sigma_z^{(1)}\rangle\langle\sigma_z^{(2)}\rangle.$$

What are the possible values of the correlation for entangled states in general. Can you use it as an entanglement test?

P4-3. Show that the trace of a product of two matrices does not depend on their order of multiplication. In other words,

$$\text{Tr}(AB) = \text{Tr}(BA),$$

even if A and B do not commute, i.e.,

$$[A, B] \equiv AB - BA \neq 0.$$

P4-4. Derive the uncertainty principle for the observables A and B for any state ψ :

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

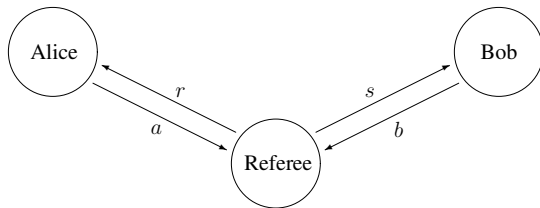
from Cauchy-Schwarz inequality

$$|\langle \phi_1 | \phi_2 \rangle|^2 \leq \langle \phi_1 | \phi_1 \rangle \langle \phi_2 | \phi_2 \rangle.$$

Hint: Select $|\phi_1\rangle = (A - \langle A \rangle) |\psi\rangle$ and $|\phi_2\rangle = (B - \langle B \rangle) |\psi\rangle$.

Lab 4 Application: The CHSH Game

- In the CHSH game, two players Alice and Bob communicate only with a referee, who randomly select a single-bit question for each player. The players receive the question (r, s) and are required to answer with a single bit. The referee then checks the answers (a, b) of Alice and Bob and determine whether the players win or lose.



- The winning conditions are $a \oplus b = r \wedge s$, i.e., $a \oplus b = 0, 0, 0, 1$, respectively, for questions $rs = 00, 01, 10, 11$.
- Can Alice and Bob win every time?
 - Let us write a_r as the answer of Alice for question r and, similarly, b_s as the answer of Bob for question s .
 - The winning conditions can be expressed as

$$a_0 \oplus b_0 = 0, \quad a_0 \oplus b_1 = 0, \quad a_1 \oplus b_0 = 0, \quad a_1 \oplus b_1 = 1.$$

- It is not possible to win every time as adding the four equations modulo 2 gives $0 = 1$.

- What is the maximum probability of winning the game using a classical strategy?
- What if Alice and Bob use a probabilistic, instead of deterministic, strategy?
- Can they win with a higher probability with a quantum strategy?
 - That is, Alice and Bob are allowed to share an entangled state and to perform quantum measurement to determine their answers to the referee's questions.

- The CHSH game is a nonlocal game that allow players to use correlations between their quantum measurement outcomes to win the game.
 - In the physical interpretation of nonlocal games, the bounds on classical strategies (allowing local hidden variables) are typically known as *Bell inequalities*. In the CHSH game, the inequality is also known as the *CHSH inequality*.
 - Quantum strategies, however, can beat the classical bounds. They are said to *violate* Bell inequalities. John Clauser experimentally tested the CHSH inequality in 1972, yielding results that violated classical limits and supported quantum mechanics. He won the Nobel Prize in Physics 50 years later for first playing the nonlocal game.